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A Novel Geophysical Approach to Investigate Mixed Wettability Effects Using Spectral Induced Polarization (SIP)

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Abstract

Conventional techniques for measuring wettability, such as contact angle measurements, Amott-Harvey, and USBM, are often time-intensive, require polished surfaces, or lack scalability. Advanced methods like Nuclear Magnetic Resonance (NMR) and dielectric techniques provide non-destructive alternatives but face challenges such as limited penetration depth and high frequency ranges (MHz–GHz), potentially overlooking key polarization mechanisms relevant for fluid-rock interaction. These limitations hinder the comprehensive characterization of reservoir properties, underscoring the need for innovative approaches. In this study, we propose a novel geophysical method known as Spectral Induced Polarization (SIP), as a promising approach to overcome these challenges in the petroleum field. The SIP technique is established for near-surface environmental applications but has limited exposure in the oil and gas domain. SIP offers a broader frequency range (1 mHz to 10 kHz) which improves the depth of investigation and is also proven to amplify the impact of fluid-rock interaction through the electrical double layer. We conducted a well-controlled experiment combining water-wet components (sand) with oil-wet particles (pyrite after aging with crude oil). Complex conductivity measurements through SIP were conducted before and after introducing the oil-wet component at full brine saturation. The results revealed significant impacts of wettability changes on SIP responses. At full brine saturation, the presence of an oil-wet component decreased conductivity throughout the frequency range compared to the water-wet test, where an increase in the phase-shift characteristic frequency and magnitude was observed in the presence of oil-wet pyrite particles due to the discontinuities of the electrical conductivity pathway and impact on the polarization mechanisms that were amplified with the presence of oil component. These findings highlight the potential

of SIP tool to evaluate the wettability changes. This novel application of SIP provides fundamental insights into wettability changes, leveraging the tool's sensitivity to fluid-solid interactions. By integrating geophysical techniques with controlled laboratory experiments, this study lays the groundwork for using SIP to characterize wettability changes more effectively. These findings pave the way for developing a SIP logging tool or incorporating SIP sensors in petrophysics laboratory setups (like core flooding). This could revolutionize industry practices by providing a robust, field-deployable solution for reservoir characterization and enhanced oil recovery strategies.

Introduction

Wettability is a key factor influencing fluid behavior in porous media, as it governs how fluids interact with solid surfaces (Abdallah et al. 2007). Understanding wettability requires knowledge of surface tension, the energy needed to increase the surface area of a single phase, and interfacial tension, the energy required to enlarge the interface between two immiscible fluids. These concepts form the basis for different wetting systems (Fig. 1): water-wet, oil-wet, and neutral-wet. In each case, the system is defined by the fluid that preferentially coats the solid surface, affecting fluid distribution and flow.

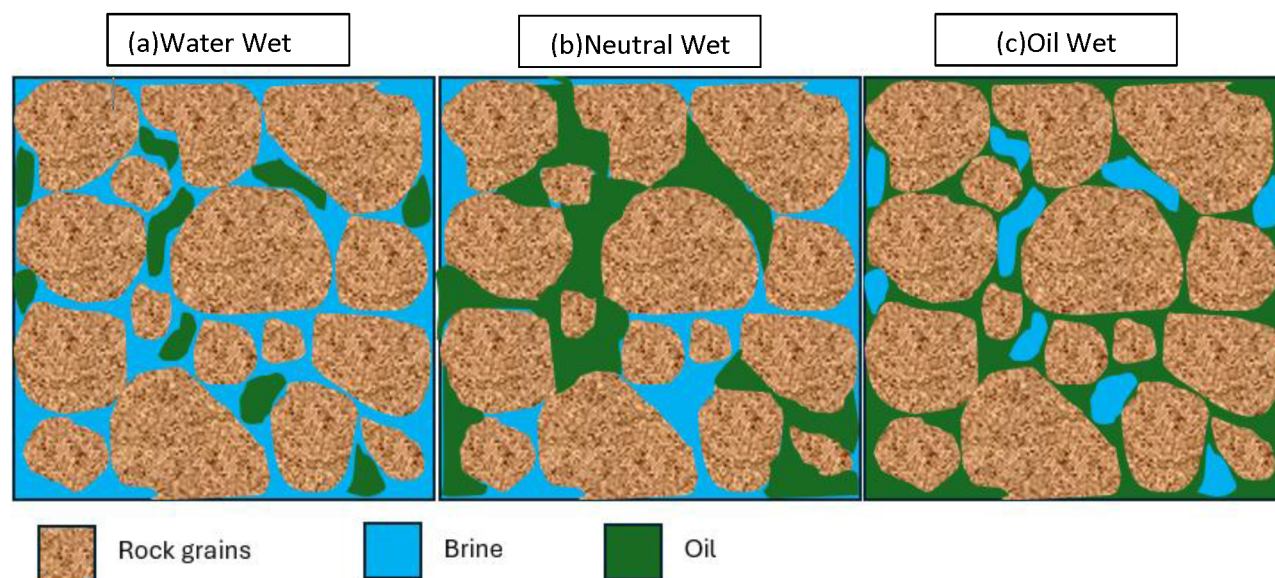


Figure 1—Various types of wettability systems: (a) illustrates a water-wet system, where water coats the grain surfaces; (b) shows a neutral wet system, with grain surfaces partially coated by both oil and water; and (c) represents an oil-wet system, where oil serves as the primary coating fluid on the grains.

In the oil and gas industry, wettability takes on paramount importance. It serves as a fundamental property for studying fluid/rock interactions and gaining insights into fluid flow dynamics. Moreover, it directly influences capillary pressure, a critical determinant in fluid displacement within reservoirs. Understanding reservoir heterogeneities in wettability becomes helpful in designing effective management strategies and optimizing hydrocarbon recovery through Enhanced Oil Recovery (EOR) techniques (Mohamed et al., 2015). Wettability's influence extends beyond the realms of the oil and gas sector into environmental domains (Deng et al, 2018), where it becomes crucial for studying contamination effects and plays a pivotal role in the context of CO₂ energy storage. (Ibrahim, A.F., 2023)

Several lab-scale techniques exist for estimating wettability, each posing its own set of challenges being very difficult to scale up with field measurements. Contact angle measurements (Anderson., 1986, Huhtamäki et al, 2018), for instance, can be difficult due to the requirement for a polished surface. Methods like Amott Harvey and USBM (Anderson., 1986, McPhee et al., 2015), which require bulk

fluid for wettability index calculations, are time-consuming and may take days to months to yield results. Recognizing these challenges, considerable efforts have been made to develop non-destructive methods applicable at both lab and field scales, such as Nuclear Magnetic Resonance (NMR) (Al-Garadi et al., 2022) and Dielectric methods (Oshaish et al., 2025). However, limitations arise in terms of penetration depth, rendering them less representative of reservoir properties. Adding to the complexities, the Dielectric method presents limitations, primarily stemming from the complex lab setup required for measurements. Moreover, the method operates within a specific frequency range, limited to higher frequencies ranging from MHz to GHz. This constrained frequency range poses a challenge as it might lead to overlooking certain polarization mechanisms critical for a comprehensive understanding of the system (Fig. 2). The subtleties and potential oversights at these higher frequencies highlight the need for alternative methods with broader applicability to capture a more complete picture of fluid-rock interactions.

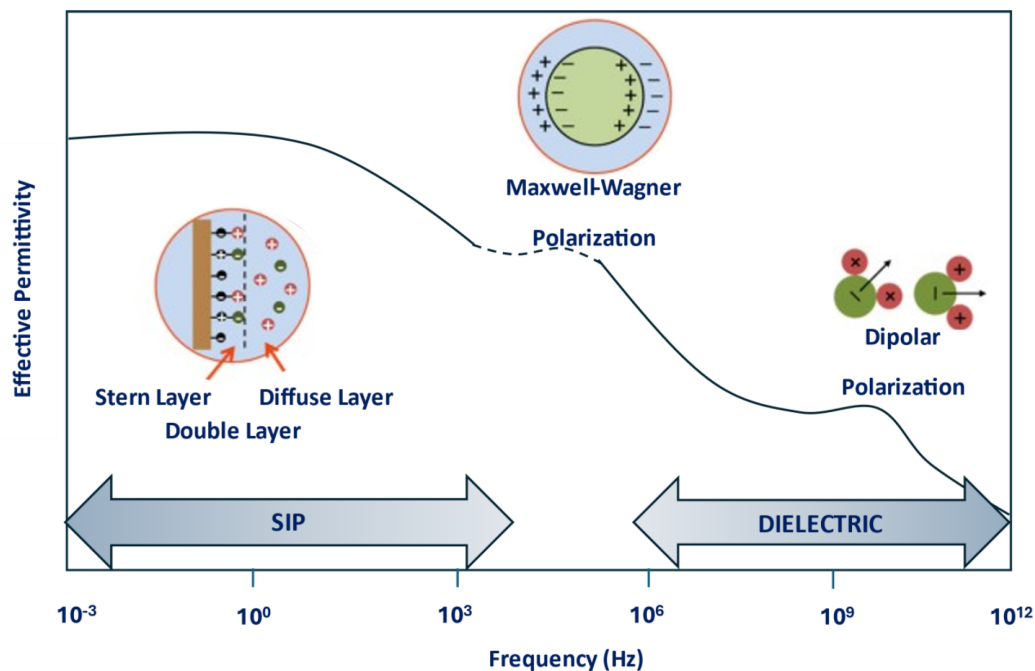


Figure 2—Effective permittivity over a range of frequency from mHz to GHz emphasizing on the dominant polarization mechanisms for both SIP and Dielectric Spectral Induced Polarization (SIP) effectively captures a range of polarization mechanisms, including the Electrical Double Layer (EDL) and Maxwell-Wagner effects, offering deep insights into interfacial behavior. While dielectric methods highlight dipolar and Maxwell-Wagner polarization, (Modified after Oshaish et al., 2023 and Binley et al., 2020)

Given these challenges, SIP stands out as a promising technique due to its polarization mechanisms such as electrical double layer (EDL) and Maxwell Wagner, and therefore its sensitivity to fluid-rock interaction. SIP is a well-established, non-destructive geophysical method traditionally applied in near-surface investigations to image subsurface conductivity and detect chargeability. Its strength lies in its ability to capture both conductive and capacitive behaviors, providing insights into a wide range of petrophysical and environmental properties. Over the years, SIP has been successfully applied across various domains. In mineral exploration, it has been used to detect disseminated sulfides and map alteration zones (Pelton et al., 1978; Mahan et al., 1986; Gurin et al., 2013). In hydrogeological studies, it has helped identify water-bearing formations and saturation changes (Seigel et al., 2007; Yehui et al., 2021; Wang et al., 2022). Additionally, SIP has been widely employed in environmental investigations, including contaminant mapping, monitor biogeochemical activity and characterizing soil and waste properties (Slater & Lesmes, 2002; Revil et al., 2012; Abdel Aal & Atekwana, 2014; Kirmizakis et al., 2020; Martin et al., 2021). More recently, SIP has gained attention in the oil and gas industry and petrophysical studies, where it is being explored as a

novel approach to characterize water saturated grains, permeability, pore size distribution, and hydrocarbon detection (Leroy et al., 2008; Revil & Florsch, 2010; Niu & Zhang, 2018; Kellal et al., 2024, Kirmizakis et al., 2024). The complex conductivity response, particularly its frequency dependence, provides insight into pore geometry, fluid distribution, and interfacial properties, offering a unique window into subsurface systems that conventional measurements alone cannot resolve.

When it comes to wettability in SIP, research remains limited, particularly regarding the influence of oil-wet components on complex conductivity and phase shift behavior. One of the earliest studies addressing this topic was conducted by Revil et al. (2011), who investigated the SIP response in oil-wet versus water-wet systems. This was later followed by Abdel Aal & Atekwana (2014), who expanded the analysis by testing both fresh oil and biodegraded oil in water-wet and oil-wet systems using unconsolidated sand. Their results have shown an increase in resistivity for the water wet sample using unconsolidated sand with oil saturation, a decrease of resistivity for the oil wet sample with an increase of oil saturation and an increase of resistivity for both oil water wet and oil wet systems, respectively, emphasizing a contradiction on the behavior of resistivity with the increase of oil saturation for both systems. More recently, Luo et al. (2022) revisited this issue using consolidated rock samples of varying mineralogical composition. Their findings showed a consistent increase in resistivity with increasing oil saturation for both water-wet and altered-wettability samples. However, the magnitude of the resistivity response was lower in oil-wet samples compared to water-wet ones. Despite these advancements, the behavior of systems with neutral wettability, commonly encountered in oil and gas reservoirs remains largely unexplored. Given the ambiguity and contradictory trends in previous studies, this work aims to investigate, for the first time, the SIP response under neutral-wet conditions, providing new insights into this overlooked but highly relevant wettability state.

SIP Background

Induced polarization (IP) and spectral induced polarization (SIP) are valuable geophysical methods for characterizing subsurface porous media due to their sensitivity to surface properties and pore structure. These techniques are especially useful in identifying geological features like mineral deposits and aquifers (Gurin et al. 2018; Revil et al. 2022; Lesmes and Frye 2001). SIP extends traditional IP by measuring frequency-dependent electrical responses, allowing for the calculation of complex conductivity or resistivity components (Binley 2015). The key measured parameters in SIP are the conductivity magnitude (σ^*) (Eq. 1) and the phase shift (ϕ) between the injected current and the resulting voltage, from which the real (σ') and imaginary components (σ'') of conductivity are derived.

$$\sigma^* = |\sigma(\omega)| e^{i\phi(\omega)} = \sigma'(\omega) + i\sigma''(\omega) \quad (1)$$

where $i^2 = -1$, $\omega = 2\pi f$ is the angular frequency, ($|\sigma(\omega)|$) is the magnitude of conductivity, and (ϕ) is the phase shift between the injected current and the measured voltage.

$$\phi(\omega) = \tan^{-1} \frac{\sigma''(\omega)}{\sigma'(\omega)} \quad (2)$$

$$|\sigma| = \sqrt{\sigma'^2 + \sigma''^2} \quad (3)$$

The real part of conductivity primarily reflects ohmic conduction, which is governed by porosity, mineral composition, fluid content, and contribution from the electrical double layer (EDL) (Ponziani et al. 2012; Weller and Slater 2015; Revil and Gresse 2021). In contrast, the imaginary component, associated with capacitive behavior, is influenced by factors such as mineral surface area, EDL thickness, and the presence of polarizable minerals (Siddiq et al. 2021; Badhafer et al. 2023). Below 1 MHz, several polarization mechanisms are active (Kemna et al. 2012). These include electrode polarization in conductive minerals like pyrite and graphite; membrane polarization arising from ion mobility restrictions in narrow pore throats

(Hördt et al. 2016); and EDL polarization at solid-fluid interfaces, involving the interaction of Stern and diffuse layers (Revil and Florsch 2010; Kemna et al. 2012). Additionally, Maxwell-Wagner polarization, which dominates at higher frequencies (>1 kHz), is driven by contrasts in electrical conductivity and permittivity between different phases within the pore structure.

Methodology

In this study, we used unconsolidated Ottawa sand with grain sizes ranging from 300 to 600 μm to represent the water-wet component in our SIP experiments. These sand samples had a porosity of approximately 50%, confirmed using Nuclear Magnetic Resonance (NMR). To introduce an oil-wet component, pyrite was ground and sieved to match the sand's grain size range. Pyrite naturally exhibits oil-wet behavior in the presence of both oil and water (Lindgren et al, 2009). To enhance its oil-wet characteristics, we aged the pyrite in Arab Light crude oil for over a month, promoting asphaltene deposition on the grain surfaces (Mohammed et al, 2021).

To investigate the effect of mixed-wettability systems on SIP response, where large pores might contain both water-wet and oil-wet components, we conducted a series of SIP measurements for comparison. These included baseline or reference tests on 100% water-wet Ottawa sand and pyrite, followed by the addition of 5%, 10%, and 20% water-wet pyrite to sand. To understand the impact of having a neutral wetting system, where a portion of the system is oil wet and the remaining is water wet, the same proportions were tested using oil-coated pyrite with water wet Ottawa sand. All samples were fully saturated with brine with a conductivity of 325 $\mu\text{S}/\text{cm}$.

SIP Measurements were performed in a frequency ranges from 0.1 mHz to 10 kHz with seven measurements per logarithmic cycle. The samples were packed in cylindrical acrylic columns, and Ag-AgCl electrodes were installed at both ends to inject current. Five potential electrodes (P1–P5) were symmetrically distributed along the column body (Fig. 3) and positioned away from the current path to minimize the influence of electrode polarization, following the recommendations of Vanhala and Soininen (1995). Several current-limiting resistors were tested to optimize signal-to-noise ratio, with 100 Ω proving most effective for the brine-saturated samples. To ensure data reliability, measurements were taken across three channels to assess repeatability and detect potential artifacts. The results presented in this study correspond to channel 3, which yielded the most homogenous and representative dataset. All tests were conducted in a temperature-controlled lab environment at $21 \pm 1^\circ\text{C}$.

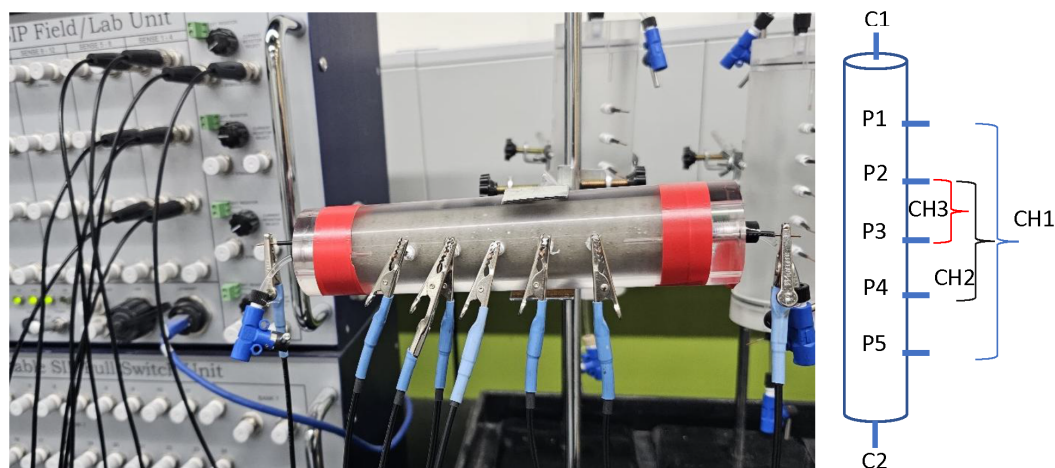


Figure 3—SIP lab scale measurement, illustrating electrodes for current injection (C1-C2) and potential electrodes (P1-P5), emphasizing different channels configuration (right). The image (left) represent an unconsolidated sample as an example of Ottawa sand with 20% water wet pyrite, fully saturated with brine.

Results and Discussion

The real conductivity and the phase shift of the water wet Ottawa sand fully saturated with brine (Fig. 4) shows a continuous behavior of about 85 $\mu\text{S}/\text{cm}$ and 0.18 mRad with a small dispersion at higher frequencies reaching up to 13.7 mRad at 10 kHz, however the real conductivity of the pyrite at its water wet state fully saturated with brine shows a continuous constant values of 1300 $\mu\text{S}/\text{cm}$ at frequencies lower than 10 Hz with a positive dispersion reaching up to almost 8000 $\mu\text{S}/\text{cm}$ at 10kHz showcasing the effect of the significant surface conductivity of the pyrite at higher frequencies due to metallic very conductive characteristic. Similarly, the phase shift changes with the frequency range giving a peak value of around 450 mRad at the characteristic frequency of around 630 Hz. The characteristic frequency for the case of pyrite is more contributed to the effect of electrical double layer (EDL) polarization giving more information about the relaxation time of the pyrite with the selected grain size.

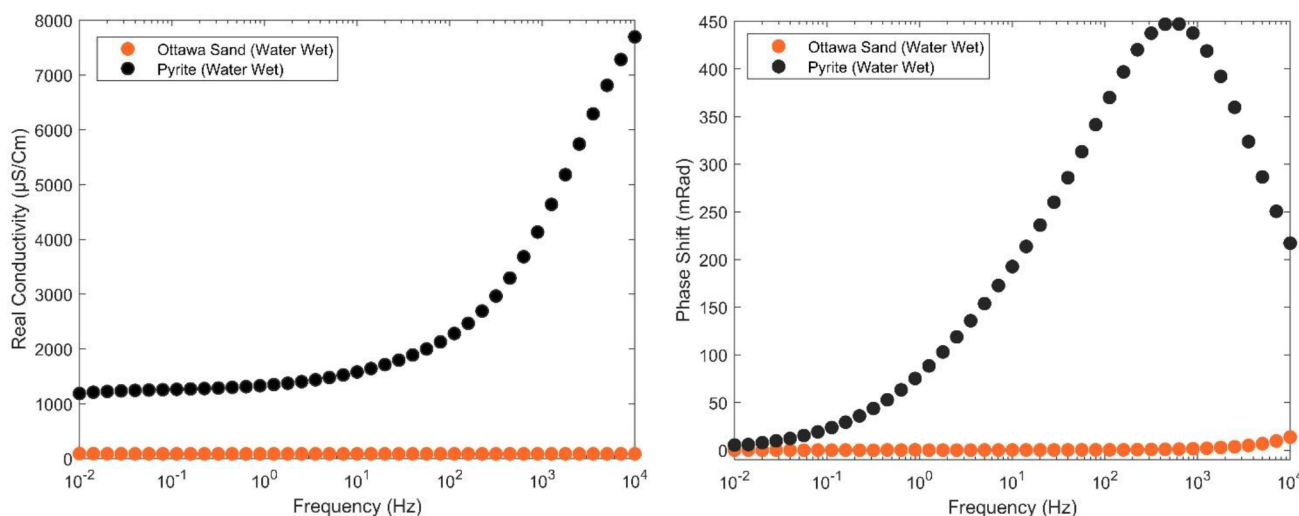


Figure 4—Real conductivity (left) and phase shift (right) over the frequency range (1mHz-10kHz) showing both SIP signatures for Ottawa sand (orange) and pyrite (black) at their water wet state fully saturated with brine as end members.

The addition of different percentages of water wet pyrite component to the water wet Ottawa sand fully saturated with brine shows an increase of conductivity with 5% pyrite reaching up to 210 $\mu\text{S}/\text{cm}$ with a very small positive dispersion at higher frequencies reaching to 230 $\mu\text{S}/\text{cm}$ (Fig. 5). Similarly, the addition of 10% and 20% pyrite particles shows a major increase of conductivity with a value of 300 $\mu\text{S}/\text{cm}$ and 500 $\mu\text{S}/\text{cm}$ respectively with a more pronounced positive dispersion at higher frequencies reaching up to 700 $\mu\text{S}/\text{cm}$ for the case of 20% pyrite. However, for the case of the phase shift, the characteristic frequency is more noticeable at higher frequencies by showing a slight shift toward the right starting from 224 Hz, 316 Hz, 447 Hz representing 5%, 10% and 20% pyrite addition to reach around 630 Hz for the 100% pyrite fully saturated with brine (Fig. 5). This indicates the gradual impact of the pyrite inclusion on the Ottawa sand dominating its signal behavior by having both an increase of the electrical real conductivity and marking the surface conductivity at higher frequencies with increase of its characteristic frequencies above 200Hz.

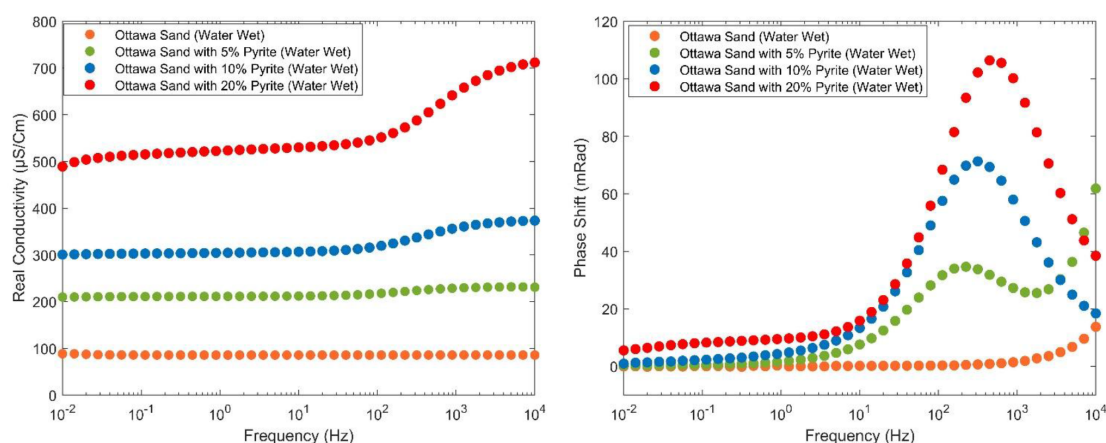


Figure 5—Real Conductivity (left) and phase shift (right) over the frequency range (1mHz-10kHz) representing SIP signatures for Ottawa sand end member (orange), Ottawa sand with 5% pyrite (green), Ottawa sand with 10% pyrite (blue) and Ottawa sand with 20% pyrite (red) all in water wet state fully saturated with brine.

As illustrated in Fig. 6, there has been a slight increase of conductivity with addition of oil coated pyrite to Ottawa sand compared to full water wet pyrite addition and the reference sample of Ottawa sand. This increase translated to 20 $\mu\text{S}/\text{cm}$, 25 $\mu\text{S}/\text{cm}$ and 65 $\mu\text{S}/\text{cm}$ rise with the addition of 5%, 10% and 20% oil coated pyrite respectively due to the change of the wettability. Similarly, there is an increase of phase shift magnitude of the same proportions beyond 10 Hz raising the impact of the oil and amplifying the discontinuities of the electrical conductivity. In order to properly compare the effect of the mixed wettability system, we are displaying the real conductivity and phase shift of 10% and 20% both water wet and oil coated pyrite with Ottawa sand Fig. 7. Clearly, we notice the big drop of the real conductivity from 10% and 20% water wet pyrite with water wet Ottawa sand to the same percentages of the neutral system (oil wet pyrite with water wet sand). This obvious decrease is related to the resistive component of oil as a film around the pyrite grains adding some discontinuities and hindering the current to pass easily compared to the water coating all grains surfaces this includes both Ottawa sand and pyrite. However, for the phase shift, there is a significant increase of magnitude at higher frequency, where the characteristic frequency is shifted toward the right due to the presence of oil and wettability alteration of pyrite component.

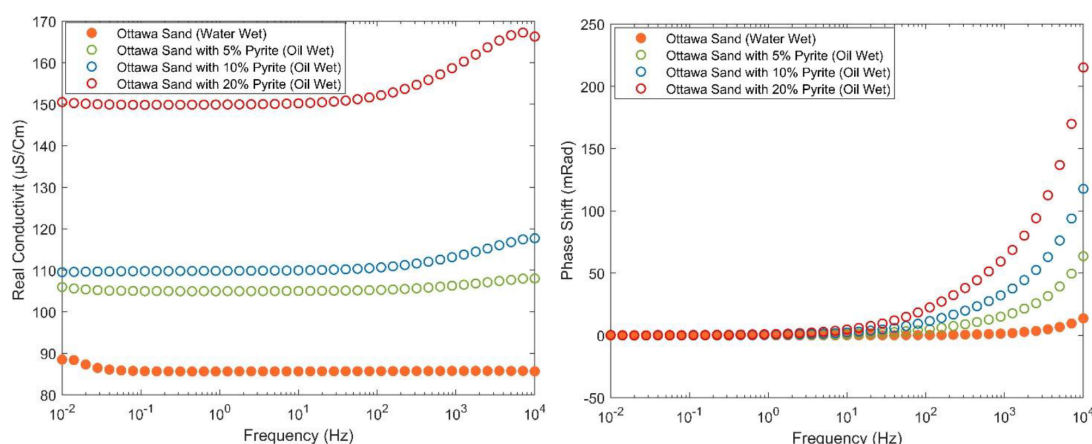


Figure 6—Real Conductivity (left) and phase shift (right) over the frequency range (1mHz-10kHz) representing SIP signatures for Ottawa sand end member (orange), Ottawa sand with 5% pyrite (green), Ottawa sand with 10% pyrite (blue) and Ottawa sand with 20% pyrite (red) all in a neutral wet (water wet sand and oil wet pyrite) state fully saturated with brine.

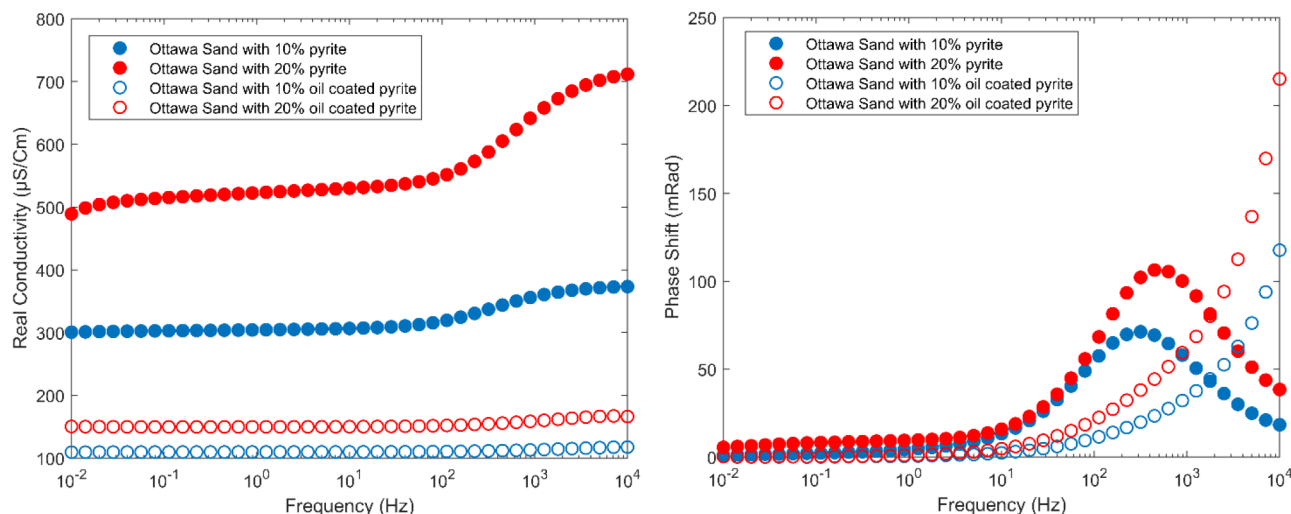


Figure 7—Real Conductivity (left) and phase shift (right) over the frequency range (1mHz-10kHz) representing SIP signatures comparing, Ottawa sand with 10% pyrite (filled-blue circles) and Ottawa sand with 20% pyrite (filled-red circles) all in water wet state fully saturated with brine with Ottawa sand with 10% oil coated pyrite (blue circles) and Ottawa sand with 20% oil coated pyrite (red circles) in a neutral wet system fully saturated with brine.

From all results shown, we can state that the increase of real conductivity and phase shift with the addition of water wet pyrite to the water wet Ottawa sand can be explained by the high surface reactivity and metallic conductivity of pyrite, which contributes significantly to the conductivity and polarization effects more importantly, the electrical double layer, where actually the pyrite particles increase the number of conductive paths within the sample, enhancing the overall conductivity of the medium. Adding to that, when the external electric field is applied, charges accumulate at the solid-fluid interface of the pyrite particles, resulting in an increased polarization therefore, higher phase shift in SIP measurement, furthermore, the higher increase of the pyrite concentration, the stronger polarization response and higher conductivity and phase angles noticed. In contrary, when adding oil coated pyrite to water wet Ottawa sand representing a more neutral wet system, can lead to a decrease of conductivity and an increase of phase magnitude with a clear shift of the characteristic frequency towards the right, while coating the pyrite with an insulating oil film, the effective surface area for the electrical conduction is reduced impacting the direct electrical pathways between the conductive pyrite particles and surrounding fluid. Even though pyrite itself is conductive, the presence of oil as an insulating barrier limits the charge transfer, resulting in lower real conductivity compared to non-coated pyrite at the same percentage. The resistive oil layer around the pyrite grains also affects the polarization mechanisms, such as the electrical double layer, where the interfacial charge build-up can be dampened, leading to the phase angle peak shifting towards higher frequencies, where less efficient charge build-up requires higher frequency to observe a comparable polarization response impacting the characteristic relaxation time of the system.

Conclusion

This work introduces a novel approach using SIP to investigate, for the first time, the impact of a neutral wetting system on complex conductivity and phase shift behavior. To achieve this work, Ottawa sand was selected to represent a water-wet component, while pyrite served as proxy for oil-wet component behavior. Numerous tests were conducted on SIP, by firstly test end members of both Ottawa sand and pyrite in their water wet state at full brine saturation. Subsequently, different proportions of pyrite (5%, 10%, 20%) were added to the Ottawa sand matrix. These composite mixtures were then tested in their native (water-wet) state and after aging with oil to simulate an oil-wet state surface. The results showed an increase of the real conductivity over the target range of frequency emphasizing the impact of the pyrite as conductive metallic material, also impacting the phase shift by amplifying the effect of polarization, especially the

electrical double layer leading to an increase of magnitude and more pronounced behavior or EDL showed as a magnitude peak at a characteristic frequency.

However, compared to full water wet system of Ottawa sand, the addition of oil coated pyrite percentages has shown a small increase of conductivity with a higher increase of magnitude of the phase shift at frequencies above 10 Hz. Focusing on the same percentages addition for both water wet pyrite and oil wet pyrite setting both full water wet and a neutral wetting systems, we notice a clear decrease of conductivity due to the presence of a resistive oil component coating the pyrite grains acting as current barriers limiting the passage of the electrical conductivity. We also observed the increase of the phase shift magnitude with a shift of the phase towards the right indicating the change of the wettability system from water wet to a neutral wet system, where it is believed that the oil coating component has an impact on the interfacial interaction between grains and fluids, results in a change of the characteristic relaxation time or frequency. These promising results shows the sensibility of SIP to any fluid-rock interaction leading to different polarization mechanisms including the EDL. For future research, we aim to expand this study by systematically comparing oil-wet and water-wet components across a broader range of mineralogy, as well as investigating the behavior of fractional and mixed-wet systems, which are commonly encountered in hydrocarbon reservoirs. Additional to further interpret and validate the observed SIP responses, we plan to develop numerical models that simulate the impact of wettability on complex conductivity.

Finally, we believe that SIP has the potential to overcome above mentioned challenges, including depth of investigation issues, providing a more comprehensive understanding of reservoir properties. This positions SIP as a fitting candidate for addressing the limitations posed by existing techniques and advancing our capabilities in wettability assessment across various scales, thereby opening new frontiers in fluid dynamics research and applications.

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